AI ASSISTED CODING
ASSIGNMENT - 14.1
ROLL NO. 2303A510J0

Task 1 – Portfolio Website Design
You are building a personal portfolio website to showcase your work.
Requirements:
• Create sections for About Me, Projects, and Contact.
• Use AI to:
o Suggest color palettes and typography.
o Create a responsive layout with Grid/Flexbox.
o Add smooth scrolling navigation.

Prompt:

I am creating a simple personal portfolio website using HTML, CSS and a little JavaScript.

Please help me build a responsive webpage that includes the following sections:

- About Me
- Projects
- Contact

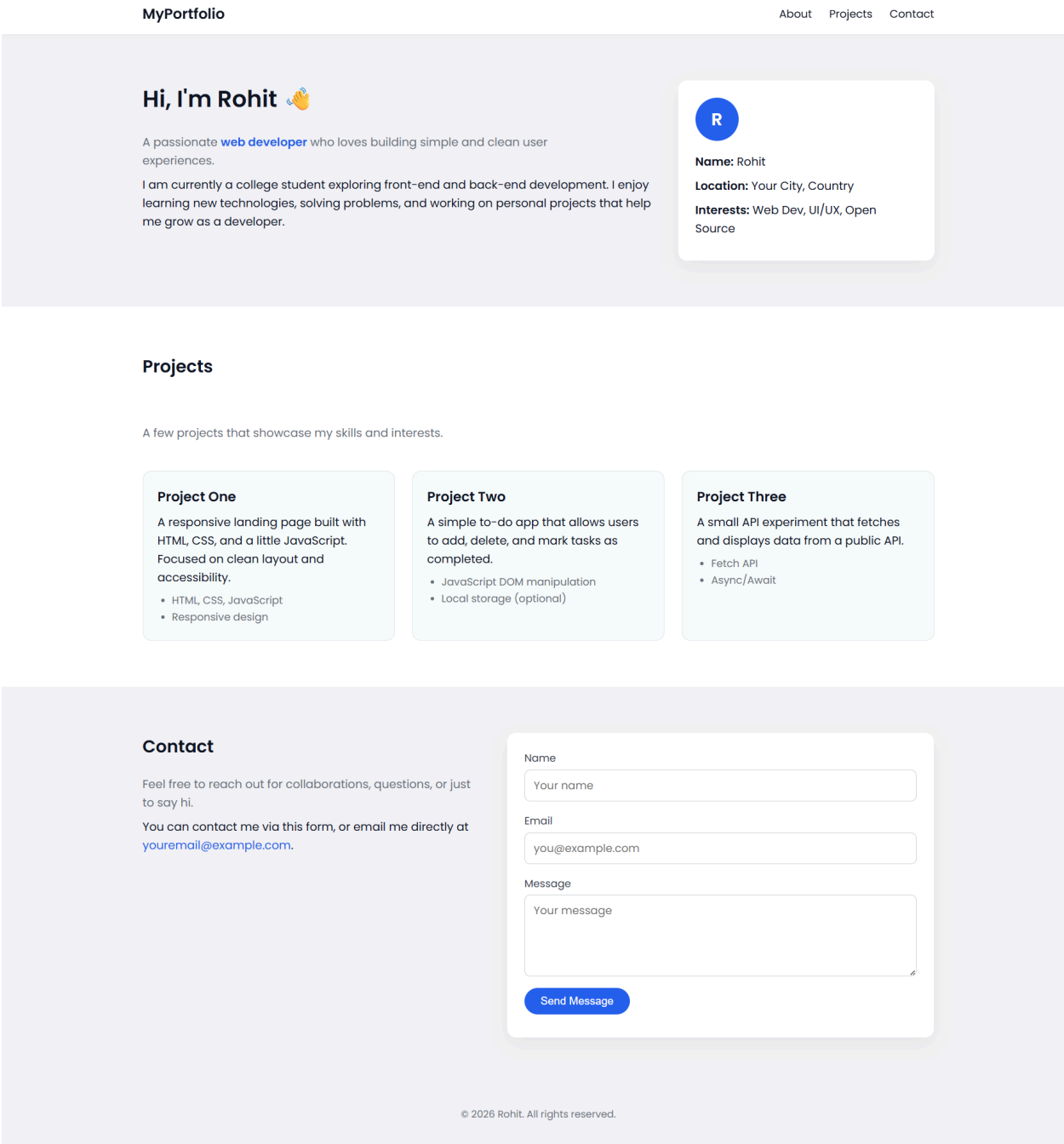
Requirements:

1. Suggest a simple color palette and good typography for a portfolio site.
2. Create a responsive layout using CSS Flexbox or Grid.
3. Add a navigation bar that scrolls smoothly to each section.
4. Keep the code clean and easy to understand since this is for a college assignment.

5. Provide HTML, CSS, and JavaScript in separate sections with comments explaining important parts.

Code is too big

Output:



Task 2 – Online Store Product Page
Design a product display page for an online store.
Requirements:
• Display product image, title, price, and "Add to Cart" button.
• Use AI to:
o Style with BEM methodology.
o Make layout responsive.
o Add hover effects and "Add to Cart" alert.

Prompt:

I am designing a simple product page for an online store using HTML, CSS, and JavaScript.

The page should display:

- Product image
- Product title
- Product price
- "Add to Cart" button

Requirements:

1. Use the BEM naming methodology for CSS classes.
2. Make the layout responsive for mobile and desktop.
3. Add hover effects for the product card and button.
4. When the "Add to Cart" button is clicked, show a JavaScript alert saying "Product added to cart".
5. Keep the code simple and well commented so it is easy to understand.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
```

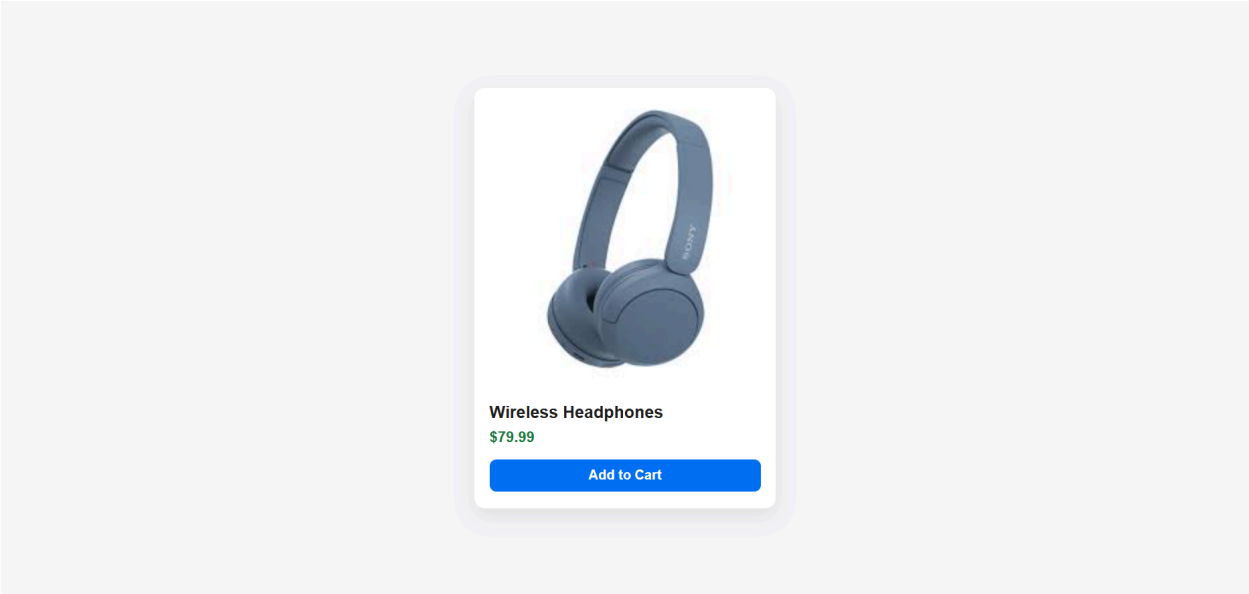
```html
<title>Simple Product Page</title>
<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<style>
  /* ========== Page Base Styles ========== */
  * {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
  }

  body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f5f5f7;
    color: #222;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    padding: 16px;
  }

  /* ========== Product Card (BEM: product-card) ========== */
  .product-card {
    background-color: #ffffff;
    border-radius: 12px;
    box-shadow: 0 10px 25px rgba(0, 0, 0, 0.08);
    max-width: 360px; /* desktop width */
    width: 100%;
    overflow: hidden;
    transition: transform 0.2s ease, box-shadow 0.2s ease;
  }

  /* Card hover effect */
  .product-card:hover {
    transform: translateY(-4px);
    box-shadow: 0 14px 30px rgba(0, 0, 0, 0.12);
  }

  /* ========== Product Image ========== */
```

```
.product-card__image-wrapper {
  width: 100%;
  overflow: hidden;
}

.product-card__image {
  display: block;
  width: 100%;
  height: auto;
  object-fit: cover;
}

/* ========== Product Content (title, price, button) ========== */
.product-card__content {
  padding: 16px 18px 20px;
}

.product-card__title {
  font-size: 1.25rem;
  font-weight: 600;
  margin-bottom: 8px;
}

.product-card__price {
  font-size: 1.1rem;
  font-weight: 700;
  color: #1a7b3f; /* green for price */
  margin-bottom: 16px;
}

/* ========== Add to Cart Button ========== */
.product-card__button {
  display: inline-block;
  width: 100%;
  text-align: center;
  padding: 10px 16px;
  background-color: #0070f3;
  color: #ffffff;
  border: none;
  border-radius: 8px;
```

```
      font-size: 1rem;
      font-weight: 600;
      cursor: pointer;
      transition: background-color 0.2s ease, transform 0.2s ease,
        box-shadow 0.2s ease;
    }

    /* Button hover effect */
    .product-card__button:hover {
      background-color: #0056c2;
      transform: translateY(-1px);
      box-shadow: 0 6px 14px rgba(0, 0, 0, 0.18);
    }

    .product-card__button:active {
      transform: translateY(0);
      box-shadow: none;
    }

    /* ========== Responsive Adjustments ========== */
    /*
      On very small screens, we already use full width with padding around
body,
      so no extra work is needed. Below is an example breakpoint for
larger
      screens if you want to tweak spacing.
    */
    @media (min-width: 768px) {
      body {
        padding: 32px;
      }
    }
  </style>
</head>
<body>
  <!--
    Product card container.
    BEM block: product-card
  -->
  <div class="product-card">
```

```html
    <!-- Image section -->
    <div class="product-card__image-wrapper">
      <!--
        Replace src with your own image path if you have one locally.
        This placeholder uses an online image.
      -->
      <img
        class="product-card__image"
        src="https://via.placeholder.com/600x400?text=Product+Image"
        alt="Sample product image"
      />
    </div>

    <!-- Text and button section -->
    <div class="product-card__content">
      <h1 class="product-card__title">Wireless Headphones</h1>
      <p class="product-card__price">$79.99</p>

      <!-- Add to cart button (BEM element: product-card__button) -->
      <button class="product-card__button" id="add-to-cart-button">
        Add to Cart
      </button>
    </div>
  </div>

  <script>
    //
    // When the "Add to Cart" button is clicked,
    // show an alert saying "Product added to cart".
    //
    const addToCartButton = document.getElementById("add-to-cart-button");

    addToCartButton.addEventListener("click", function () {
      alert("Product added to cart");
    });
  </script>
</body>
</html>
```

Output:



Task 3 – Event Registration Form
Build an event registration form for a conference.
Requirements:
• Collect name, email, phone number, and session selection.
• Use AI to:
o Add form validation with JavaScript.
o Make the form accessible with labels and ARIA.
o Style with a professional look

Prompt:

I need to build an event registration form for a conference using HTML, CSS, and JavaScript.

The form should collect:

- **Name**
- **Email**
- **Phone Number**
- **Session selection (dropdown)**

Requirements:

1. **Add JavaScript form validation for required fields and proper email format.**
2. **Make the form accessible using proper labels and ARIA attributes.**
3. **Style the form so it looks professional and clean.**
4. **Make sure the layout is responsive for smaller screens.**
5. **Include comments explaining the validation logic.**

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>Conference Event Registration</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <style>
    /* ====== Base styles ====== */
    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      font-family: system-ui, -apple-system, BlinkMacSystemFont, "Segoe
UI",
        sans-serif;
      background: #f3f4f6;
      color: #111827;
      display: flex;
      justify-content: center;
      padding: 24px;
    }

    .page-wrapper {
      max-width: 600px;
```

```
  width: 100%;
}

.card {
  background: #ffffff;
  border-radius: 12px;
  padding: 24px 24px 28px;
  box-shadow: 0 10px 30px rgba(15, 23, 42, 0.12);
}

h1 {
  margin-top: 0;
  font-size: 1.6rem;
  margin-bottom: 4px;
}

.subtitle {
  margin: 0 0 20px;
  color: #6b7280;
  font-size: 0.95rem;
}

/* ====== Form layout ====== */
form {
  margin-top: 8px;
}

.form-group {
  margin-bottom: 16px;
}

label {
  display: inline-block;
  margin-bottom: 6px;
  font-weight: 600;
  font-size: 0.95rem;
  color: #111827;
}

.required-badge {
```

```
    color: #dc2626;
    margin-left: 4px;
    font-weight: 600;
  }

  input[type="text"],
  input[type="email"],
  input[type="tel"],
  select {
    width: 100%;
    padding: 10px 12px;
    border-radius: 8px;
    border: 1px solid #d1d5db;
    font-size: 0.95rem;
    transition: border-color 0.15s ease, box-shadow 0.15s ease,
      background-color 0.15s ease;
    background-color: #f9fafb;
  }

  input:focus,
  select:focus {
    outline: none;
    border-color: #2563eb;
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.25);
    background-color: #ffffff;
  }

  .error {
    border-color: #dc2626 !important;
    background-color: #fef2f2;
  }

  .error-message {
    margin-top: 4px;
    font-size: 0.82rem;
    color: #b91c1c;
    min-height: 1.1em;
  }

  .form-footer {
```

```
  margin-top: 18px;
  display: flex;
  flex-direction: column;
  gap: 8px;
}

.submit-btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 10px 16px;
  border-radius: 999px;
  border: none;
  background: linear-gradient(135deg, #2563eb, #1d4ed8);
  color: #ffffff;
  font-weight: 600;
  font-size: 0.95rem;
  cursor: pointer;
  box-shadow: 0 10px 18px rgba(37, 99, 235, 0.35);
  transition: transform 0.1s ease, box-shadow 0.1s ease,
    background 0.15s ease;
}

.submit-btn:hover {
  transform: translateY(-1px);
  box-shadow: 0 12px 22px rgba(37, 99, 235, 0.4);
  background: linear-gradient(135deg, #1d4ed8, #1d4ed8);
}

.submit-btn:active {
  transform: translateY(0);
  box-shadow: 0 6px 10px rgba(37, 99, 235, 0.3);
}

.submit-btn:focus-visible {
  outline: 3px solid #1d4ed8;
  outline-offset: 3px;
}
```

```
    .helper-text {
      font-size: 0.8rem;
      color: #6b7280;
    }

    .success-banner {
      margin-top: 10px;
      padding: 8px 10px;
      border-radius: 8px;
      background-color: #ecfdf3;
      color: #166534;
      font-size: 0.88rem;
      border: 1px solid #bbf7d0;
      display: none;
    }

    .success-banner[aria-hidden="false"] {
      display: block;
    }

    /* ====== Responsive tweaks ====== */
    @media (max-width: 480px) {
      body {
        padding: 16px;
      }

      .card {
        padding: 18px 16px 24px;
      }

      h1 {
        font-size: 1.35rem;
      }
    }
  </style>
</head>
<body>
  <div class="page-wrapper">
    <section class="card" aria-labelledby="form-heading">
      <h1 id="form-heading">Conference Registration</h1>
```

```
      <p class="subtitle">
        Reserve your spot for the upcoming conference. Fields marked with
        <span class="required-badge" aria-hidden="true">*</span> are
required.
      </p>

      <!-- ARIA live region for form-level status messages -->
      <div
        id="form-status"
        class="success-banner"
        role="status"
        aria-live="polite"
        aria-hidden="true"
      ></div>

      <form id="registration-form" novalidate
aria-describedby="form-help">
        <p id="form-help" class="helper-text">
          Your information is used only for event communication.
        </p>

        <!-- Name -->
        <div class="form-group">
          <label for="name">
            Full Name
            <span class="required-badge" aria-hidden="true">*</span>
          </label>
          <input
            type="text"
            id="name"
            name="name"
            autocomplete="name"
            aria-required="true"
            aria-describedby="name-error"
          />
          <div
            id="name-error"
            class="error-message"
            role="alert"
            aria-live="polite"
```

```
        ></div>
      </div>

      <!-- Email -->
      <div class="form-group">
        <label for="email">
          Email
          <span class="required-badge" aria-hidden="true">*</span>
        </label>
        <input
          type="email"
          id="email"
          name="email"
          autocomplete="email"
          inputmode="email"
          aria-required="true"
          aria-describedby="email-error"
        />
        <div
          id="email-error"
          class="error-message"
          role="alert"
          aria-live="polite"
        ></div>
      </div>

      <!-- Phone -->
      <div class="form-group">
        <label for="phone">
          Phone Number
          <span class="required-badge" aria-hidden="true">*</span>
        </label>
        <input
          type="tel"
          id="phone"
          name="phone"
          autocomplete="tel"
          inputmode="tel"
          aria-required="true"
          aria-describedby="phone-error"
```

```
          />
          <div
            id="phone-error"
            class="error-message"
            role="alert"
            aria-live="polite"
          ></div>
        </div>

        <!-- Session selection -->
        <div class="form-group">
          <label for="session">
            Session Selection
            <span class="required-badge" aria-hidden="true">*</span>
          </label>
          <select
            id="session"
            name="session"
            aria-required="true"
            aria-describedby="session-error"
          >
            <option value="">-- Please choose a session --</option>
            <option value="keynote">Keynote & Opening Remarks</option>
            <option value="track-ai">AI & Machine Learning Track</option>
            <option value="track-web">Modern Web Development
Track</option>
            <option value="track-data">Data Science & Analytics
Track</option>
            <option value="workshop">Hands-on Workshop
(Afternoon)</option>
          </select>
          <div
            id="session-error"
            class="error-message"
            role="alert"
            aria-live="polite"
          ></div>
        </div>

        <div class="form-footer">
```

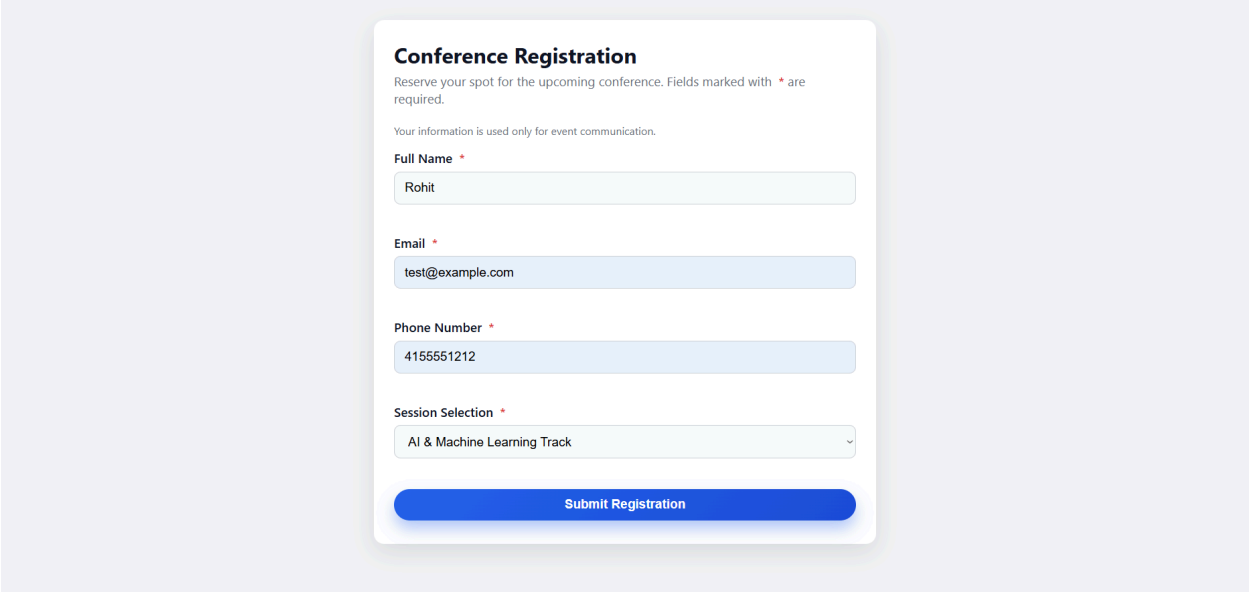
```
          <button type="submit" class="submit-btn">
            Submit Registration
          </button>
        </div>
      </form>
    </section>
  </div>

  <script>
    // ====== Simple form validation logic ======
    // We validate on submit, and also clear errors when the user starts
typing again.

    const form = document.getElementById("registration-form");
    const statusBanner = document.getElementById("form-status");

    // Helper to show an error on a field
    function showError(inputEl, errorEl, message) {
      inputEl.classList.add("error");
      inputEl.setAttribute("aria-invalid", "true");
      errorEl.textContent = message;
    }

    // Helper to clear an error on a field
    function clearError(inputEl, errorEl) {
      inputEl.classList.remove("error");
      inputEl.removeAttribute("aria-invalid");
      errorEl.textContent = "";
    }

    // Basic email pattern for client-side validation (not perfect, but
good enough)
    function isValidEmail(email) {
      const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
      return emailPattern.test(email.trim());
    }

    // Basic phone validation: at least 7 digits, ignoring spaces, dashes,
and parentheses
    function isValidPhone(phone) {
```

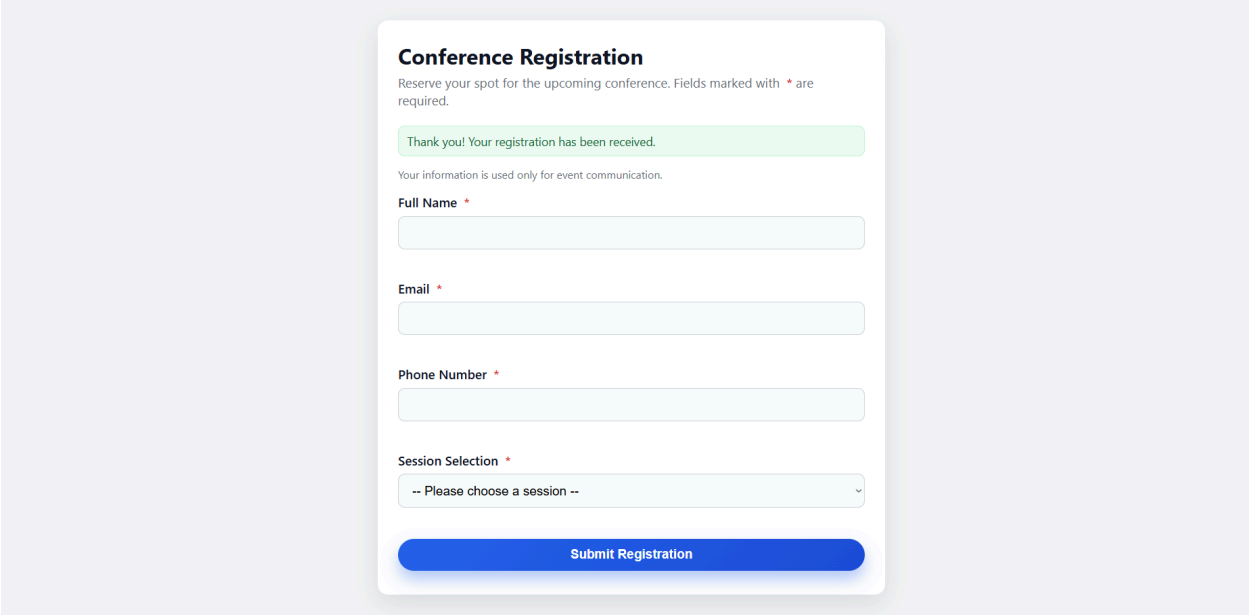
```
    const digits = phone.replace(/\D/g, "");
    return digits.length >= 7;
  }

  form.addEventListener("submit", function (event) {
    event.preventDefault(); // prevent default form submission

    // Get references to inputs and error containers
    const nameInput = document.getElementById("name");
    const emailInput = document.getElementById("email");
    const phoneInput = document.getElementById("phone");
    const sessionSelect = document.getElementById("session");

    const nameError = document.getElementById("name-error");
    const emailError = document.getElementById("email-error");
    const phoneError = document.getElementById("phone-error");
    const sessionError = document.getElementById("session-error");

    // Clear previous status message
    statusBanner.textContent = "";
    statusBanner.setAttribute("aria-hidden", "true");

    // Clear previous field errors
    clearError(nameInput, nameError);
    clearError(emailInput, emailError);
    clearError(phoneInput, phoneError);
    clearError(sessionSelect, sessionError);

    let hasError = false;

    // ==== Required field checks ====

    // Name required
    if (!nameInput.value.trim()) {
      showError(nameInput, nameError, "Please enter your full name.");
      hasError = true;
    }

    // Email required + format
    if (!emailInput.value.trim()) {
```

```
        showError(emailInput, emailError, "Please enter your email
address.");
        hasError = true;
      } else if (!isValidEmail(emailInput.value)) {
        // Email format validation using simple regex pattern
        showError(
          emailInput,
          emailError,
          "Please enter a valid email address (e.g., name@example.com)."
        );
        hasError = true;
      }

      // Phone required + basic format
      if (!phoneInput.value.trim()) {
        showError(phoneInput, phoneError, "Please enter your phone
number.");
        hasError = true;
      } else if (!isValidPhone(phoneInput.value)) {
        // Phone validation checks for at least 7 digits
        showError(
          phoneInput,
          phoneError,
          "Please enter a valid phone number with at least 7 digits."
        );
        hasError = true;
      }

      // Session selection required
      if (!sessionSelect.value) {
        showError(
          sessionSelect,
          sessionError,
          "Please select a session you would like to attend."
        );
        hasError = true;
      }

      // If any error exists, focus the first invalid field and stop
      if (hasError) {
```

```
        const firstErrorField = form.querySelector(
          ".error, [aria-invalid='true']"
        );
        if (firstErrorField) {
          firstErrorField.focus();
        }
        return;
      }

      // If no errors: simulate successful submission
      statusBanner.textContent =
        "Thank you! Your registration has been received.";
      statusBanner.setAttribute("aria-hidden", "false");

      // Optionally reset the form
      form.reset();
    });

    // Clear error message when user starts correcting a field
    ["input", "change"].forEach((eventType) => {
      form.addEventListener(
        eventType,
        function (event) {
          const target = event.target;
          if (!target.id) return;

          const errorEl = document.getElementById(target.id + "-error");
          if (errorEl) {
            clearError(target, errorEl);
          }
        },
        true
      );
    });
  </script>
</body>
</html>
```

Output:





Task 4 – Resume Website

Build a one-page online resume.

Requirements:

- **Sections: Profile Summary, Skills, Education, Experience.**
- **Use AI to:**
  - **Propose professional typography styles.**
  - **Align sections using CSS Grid.**

o Add smooth scroll navigation between sections.

Prompt:

I want to build a one-page resume website using HTML, CSS, and JavaScript.

The webpage should include these sections:

- **Profile Summary**
- **Skills**
- **Education**
- **Experience**

Requirements:

1. **Suggest professional typography suitable for a resume website.**
2. **Align and structure sections using CSS Grid.**
3. **Add a simple navigation menu that smoothly scrolls to each section.**
4. **Keep the design clean and minimal like a professional resume.**
5. **Provide clear and commented code.**

Code:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Resume - Your Name</title>

  <!--
    Typography:
    - Uses "Segoe UI" (Windows), with fallbacks to system UI fonts.
    - Clean and professional, similar to what you see in MS Office.
  -->
  <style>
    /* ====== Base Reset & Layout ====== */
```

```
*,
*::before,
*::after {
  box-sizing: border-box;
}

html {
  scroll-behavior: smooth;
}

body {
  margin: 0;
  padding: 0;
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
    "Helvetica Neue", Arial, sans-serif;
  background-color: #f5f6f8;
  color: #222;
  line-height: 1.6;
}

/* ====== Page Wrapper ====== */
.page {
  max-width: 960px;
  margin: 0 auto;
  padding: 24px 16px 40px;
}

/* ====== Header & Navigation ====== */
header {
  position: sticky;
  top: 0;
  z-index: 10;
  background-color: #ffffffee;
  backdrop-filter: blur(6px);
  border-bottom: 1px solid #e0e0e0;
}

.header-inner {
  max-width: 960px;
  margin: 0 auto;
```

```
  padding: 12px 16px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
}

.branding {
  display: flex;
  flex-direction: column;
}

.branding__name {
  font-size: 1.25rem;
  font-weight: 600;
  letter-spacing: 0.03em;
}

.branding__role {
  font-size: 0.9rem;
  color: #666;
}

/* Simple horizontal navigation */
nav ul {
  margin: 0;
  padding: 0;
  display: flex;
  list-style: none;
  gap: 16px;
  font-size: 0.9rem;
}

nav a {
  text-decoration: none;
  color: #444;
  padding: 4px 0;
  border-bottom: 2px solid transparent;
  transition: color 0.15s ease, border-color 0.15s ease;
}
```

```
nav a:hover,
nav a:focus-visible {
  color: #0057b8;
  border-color: #0057b8;
  outline: none;
}

/* Mobile nav: simple wrap */
@media (max-width: 640px) {
  .header-inner {
    flex-direction: column;
    align-items: flex-start;
  }

  nav ul {
    flex-wrap: wrap;
  }
}

/* ====== Main Layout Using CSS Grid ====== */
main {
  margin-top: 24px;
}

/*
  Use a single-column layout on small screens,
  and a two-column grid on larger screens.
*/
.grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 24px;
}

@media (min-width: 768px) {
  .grid {
    grid-template-columns: 2fr 3fr;
    align-items: start;
  }
```

```
}

/* ====== Card / Section Styling ====== */
section {
  background-color: #fff;
  border-radius: 6px;
  padding: 18px 20px;
  border: 1px solid #e0e0e0;
}

section + section {
  margin-top: 16px;
}

.section-title {
  font-size: 1.05rem;
  text-transform: uppercase;
  letter-spacing: 0.12em;
  color: #555;
  margin: 0 0 12px;
  border-bottom: 1px solid #eee;
  padding-bottom: 6px;
}

.muted {
  color: #777;
  font-size: 0.92rem;
}

/* ====== Profile Summary ====== */
#profile-summary p {
  margin-top: 4px;
  font-size: 0.98rem;
}

/* ====== Skills ====== */
.skills-list {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(120px, 1fr));
  gap: 8px 16px;
```

```
  padding-left: 0;
  list-style: none;
  font-size: 0.95rem;
}

.skills-list li::before {
  content: "•";
  color: #0057b8;
  margin-right: 6px;
}

/* ====== Education & Experience ====== */
.entry {
  margin-bottom: 14px;
}

.entry:last-child {
  margin-bottom: 0;
}

.entry__header {
  display: flex;
  justify-content: space-between;
  gap: 12px;
  font-size: 0.96rem;
  font-weight: 500;
}

.entry__sub {
  font-size: 0.9rem;
  color: #666;
  margin-top: 2px;
}

.entry__details {
  margin-top: 4px;
  font-size: 0.92rem;
}

.entry__details ul {
```

```
      margin: 4px 0 0 16px;
      padding: 0;
    }

    .entry__details li {
      margin-bottom: 2px;
    }

    /* ====== Small Footer ====== */
    footer {
      margin-top: 32px;
      text-align: center;
      font-size: 0.85rem;
      color: #888;
    }
  </style>
</head>

<body>
  <!-- ====== Sticky Header with Navigation ====== -->
  <header>
    <div class="header-inner">
      <div class="branding">
        <span class="branding__name">Your Name</span>
        <span class="branding__role">Job Title / Role</span>
      </div>
      <nav aria-label="Resume sections">
        <ul>
          <!--
            Each navigation link scrolls smoothly to its corresponding
section.
            The href value must match the section's id.
          -->
          <li><a href="#profile-summary">Profile</a></li>
          <li><a href="#skills">Skills</a></li>
          <li><a href="#education">Education</a></li>
          <li><a href="#experience">Experience</a></li>
        </ul>
      </nav>
    </div>
```

```
  </header>

  <!-- ====== Main Resume Content ====== -->
  <div class="page">
    <main class="grid">
      <!-- Left column: Profile + Skills -->
      <div>
        <!-- Profile Summary Section -->
        <section id="profile-summary">
          <h2 class="section-title">Profile Summary</h2>
          <p>
            Concise professional summary goes here. Highlight your key
strengths,
            years of experience, and the type of roles or projects you are
most
            interested in. Aim for 3–4 lines of clear, impactful text.
          </p>
          <p class="muted">
            Example: Detail-oriented software engineer with 4+ years of
experience
            building web applications using modern JavaScript frameworks
and cloud
            technologies.
          </p>
        </section>

        <!-- Skills Section -->
        <section id="skills">
          <h2 class="section-title">Skills</h2>
          <!--
            Skills are laid out using CSS Grid.
            Adjust the list items to match your actual skills.
          -->
          <ul class="skills-list">
            <li>HTML &amp; CSS</li>
            <li>JavaScript / TypeScript</li>
            <li>React / Angular</li>
            <li>Node.js</li>
            <li>REST APIs</li>
            <li>SQL / NoSQL Databases</li>
```

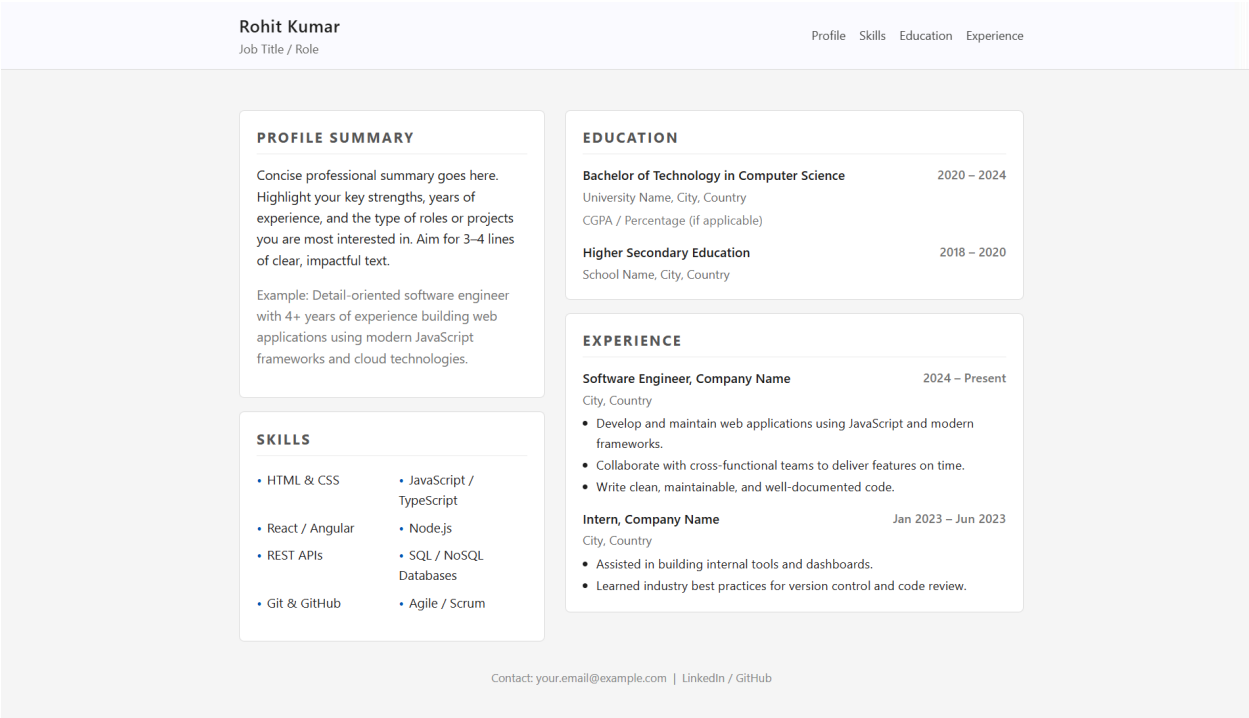
```html
        <li>Git &amp; GitHub</li>
        <li>Agile / Scrum</li>
      </ul>
    </section>
  </div>

  <!-- Right column: Education + Experience -->
  <div>
    <!-- Education Section -->
    <section id="education">
      <h2 class="section-title">Education</h2>

      <div class="entry">
        <div class="entry__header">
          <span>Bachelor of Technology in Computer Science</span>
          <span class="muted">2020 – 2024</span>
        </div>
        <div class="entry__sub">
          University Name, City, Country
        </div>
        <div class="entry__details muted">
          CGPA / Percentage (if applicable)
        </div>
      </div>

      <div class="entry">
        <div class="entry__header">
          <span>Higher Secondary Education</span>
          <span class="muted">2018 – 2020</span>
        </div>
        <div class="entry__sub">
          School Name, City, Country
        </div>
      </div>
    </section>

    <!-- Experience Section -->
    <section id="experience">
      <h2 class="section-title">Experience</h2>
```

```
          <!-- Example job entry 1 -->
          <div class="entry">
            <div class="entry__header">
              <span>Software Engineer, Company Name</span>
              <span class="muted">2024 – Present</span>
            </div>
            <div class="entry__sub">
              City, Country
            </div>
            <div class="entry__details">
              <ul>
                <li>Develop and maintain web applications using JavaScript
and modern frameworks.</li>
                <li>Collaborate with cross-functional teams to deliver
features on time.</li>
                <li>Write clean, maintainable, and well-documented
code.</li>
              </ul>
            </div>
          </div>

          <!-- Example job entry 2 -->
          <div class="entry">
            <div class="entry__header">
              <span>Intern, Company Name</span>
              <span class="muted">Jan 2023 – Jun 2023</span>
            </div>
            <div class="entry__sub">
              City, Country
            </div>
            <div class="entry__details">
              <ul>
                <li>Assisted in building internal tools and
dashboards.</li>
                <li>Learned industry best practices for version control
and code review.</li>
              </ul>
            </div>
          </div>
        </section>
```

```
      </div>
    </main>

    <footer>
      <!--
        You can add contact information or links here,
        such as email, LinkedIn, or GitHub.
      -->
      <span>Contact: your.email@example.com  |  LinkedIn /
GitHub</span>
    </footer>
  </div>

  <!-- ====== Minimal JavaScript for Nav Highlighting (Optional) ======
-->
  <script>
    // This script highlights the nav link for the section currently in
view.
    // It keeps the behavior simple and unobtrusive.

    document.addEventListener("DOMContentLoaded", function () {
      const navLinks = Array.from(document.querySelectorAll("nav a"));

      // Map section id to corresponding nav link
      const linkById = {};
      navLinks.forEach((link) => {
        const id = link.getAttribute("href").slice(1);
        linkById[id] = link;
      });

      const sections = Array.from(document.querySelectorAll("main
section"));

      function onScroll() {
        let activeId = null;
        const offset = 120; // accounts for sticky header height

        sections.forEach((section) => {
          const rect = section.getBoundingClientRect();
          if (rect.top - offset <= 0 && rect.bottom - offset > 0) {
```

```
            activeId = section.id;
          }
        });

        navLinks.forEach((link) => {
          link.style.fontWeight = "400";
        });

        if (activeId && linkById[activeId]) {
          linkById[activeId].style.fontWeight = "600";
        }
      }

      document.addEventListener("scroll", onScroll, { passive: true });
      onScroll();
    });
  </script>
</body>

</html>
```

Output:



Task 5 – Restaurant Menu Page
Design a digital menu page for a restaurant.
Requirements:
• Display menu items with name, description, and price.
• Use AI to:
o Suggest a food-friendly color palette.
o Design a responsive card-based layout.
o Add hover effects on menu items.

Prompt:

I am designing a digital restaurant menu page using HTML and CSS.

Each menu item should show:

- **Item name**
- **Short description**
- **Price**

Requirements:

1. **Suggest a color palette that fits a food or restaurant theme.**
2. **Display menu items using a responsive card-based layout.**
3. **Add hover effects on menu cards to make the design interactive.**
4. **Ensure the page works well on both mobile and desktop screens.**
5. **Keep the code simple and readable.**

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Restaurant Menu</title>
  <style>
    /* Global styles */
    :root {
      --bg-color: #fff7ec;
      --card-bg: #ffffff;
      --primary-text: #2c3e50;
      --muted-text: #7f8c8d;
      --accent: #d35400;
      --accent-light: #f0c987;
      --shadow-soft: 0 4px 12px rgba(0, 0, 0, 0.08);
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: system-ui, -apple-system, BlinkMacSystemFont, "Segoe
UI",
        sans-serif;
```

```
  background: var(--bg-color);
  color: var(--primary-text);
  line-height: 1.5;
}

/* Page layout */
.menu-page {
  max-width: 1100px;
  margin: 0 auto;
  padding: 1.5rem 1rem 3rem;
}

header {
  text-align: center;
  margin-bottom: 2rem;
}

header h1 {
  font-size: 2.2rem;
  letter-spacing: 0.05em;
  text-transform: uppercase;
  margin-bottom: 0.5rem;
}

header p {
  color: var(--muted-text);
  font-size: 0.95rem;
}

/* Menu grid */
.menu-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(230px, 1fr));
  gap: 1.25rem;
}

/* Menu card */
.menu-card {
  background: var(--card-bg);
  border-radius: 12px;
```

```
  padding: 1.1rem 1.1rem 1.2rem;
  box-shadow: var(--shadow-soft);
  border: 1px solid rgba(0, 0, 0, 0.02);
  transition:
    transform 0.18s ease,
    box-shadow 0.18s ease,
    border-color 0.18s ease,
    background-color 0.18s ease;
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  min-height: 150px;
}

.menu-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.12);
  border-color: var(--accent-light);
  background-color: #fffdf8;
}

.menu-card-header {
  display: flex;
  justify-content: space-between;
  gap: 0.75rem;
  margin-bottom: 0.4rem;
}

.menu-item-name {
  font-weight: 600;
  font-size: 1.05rem;
}

.menu-item-price {
  font-weight: 600;
  color: var(--accent);
  white-space: nowrap;
}

.menu-item-desc {
```

```
      color: var(--muted-text);
      font-size: 0.9rem;
      margin-top: 0.2rem;
    }

    /* Simple category label (optional) */
    .menu-category {
      display: inline-block;
      font-size: 0.75rem;
      text-transform: uppercase;
      letter-spacing: 0.08em;
      color: var(--accent);
      background: rgba(211, 84, 0, 0.07);
      padding: 0.25rem 0.55rem;
      border-radius: 999px;
      margin-bottom: 0.65rem;
    }

    /* Responsive tweaks */
    @media (max-width: 600px) {
      header h1 {
        font-size: 1.7rem;
      }

      .menu-page {
        padding: 1.2rem 0.85rem 2.5rem;
      }

      .menu-card {
        padding: 1rem;
      }
    }
  </style>
</head>
<body>
  <div class="menu-page">
    <header>
      <h1>Sunset Bistro</h1>
      <p>Freshly prepared dishes made with seasonal ingredients.</p>
    </header>
```

```
    <section>
      <span class="menu-category">Starters</span>
      <div class="menu-grid">
        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Bruschetta Trio</h2>
              <span class="menu-item-price">$8.50</span>
            </div>
            <p class="menu-item-desc">
              Toasted baguette topped with tomato basil, roasted peppers,
and
              whipped goat cheese.
            </p>
          </div>
        </article>

        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Creamy Tomato Soup</h2>
              <span class="menu-item-price">$6.00</span>
            </div>
            <p class="menu-item-desc">
              Slow-simmered tomatoes with basil, topped with herb
croutons.
            </p>
          </div>
        </article>
      </div>
    </section>

    <section style="margin-top: 2rem;">
      <span class="menu-category">Mains</span>
      <div class="menu-grid">
        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Grilled Lemon Herb Chicken</h2>
```

```
              <span class="menu-item-price">$16.90</span>
            </div>
            <p class="menu-item-desc">
              Marinated chicken breast with roasted potatoes and seasonal
              vegetables.
            </p>
          </div>
        </article>

        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Wild Mushroom Risotto</h2>
              <span class="menu-item-price">$15.50</span>
            </div>
            <p class="menu-item-desc">
              Arborio rice cooked in vegetable stock with wild mushrooms
and
              parmesan.
            </p>
          </div>
        </article>

        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Spicy Shrimp Pasta</h2>
              <span class="menu-item-price">$17.40</span>
            </div>
            <p class="menu-item-desc">
              Linguine tossed in a garlic-chili tomato sauce with sautéed
              shrimp.
            </p>
          </div>
        </article>
      </div>
    </section>

    <section style="margin-top: 2rem;">
      <span class="menu-category">Desserts</span>
```

```
      <div class="menu-grid">
        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Classic Tiramisu</h2>
              <span class="menu-item-price">$7.20</span>
            </div>
            <p class="menu-item-desc">
              Espresso-soaked ladyfingers layered with mascarpone cream
and
              cocoa.
            </p>
          </div>
        </article>

        <article class="menu-card">
          <div>
            <div class="menu-card-header">
              <h2 class="menu-item-name">Warm Apple Crumble</h2>
              <span class="menu-item-price">$6.80</span>
            </div>
            <p class="menu-item-desc">
              Baked apples with cinnamon crumble, served with vanilla ice
cream.
            </p>
          </div>
        </article>
      </div>
    </section>
  </div>
</body>
</html>
```

Output:

SUNSET BISTRO

Freshly prepared dishes made with seasonal ingredients.

STARTERS

Bruschetta Trio — $8.50

Toasted baguette topped with tomato basil, roasted peppers, and whipped goat cheese.

Creamy Tomato Soup — $6.00

Slow-simmered tomatoes with basil, topped with herb croutons.

MAINS

Grilled Lemon Herb Chicken — $16.90

Marinated chicken breast with roasted potatoes and seasonal vegetables.

Wild Mushroom Risotto — $15.50

Arborio rice cooked in vegetable stock with wild mushrooms and parmesan.

Spicy Shrimp Pasta — $17.40

Linguine tossed in a garlic-chili tomato sauce with sautéed shrimp.

DESSERTS

Classic Tiramisu — $7.20

Espresso-soaked ladyfingers layered with mascarpone cream and cocoa.

Warm Apple Crumble — $6.80

Baked apples with cinnamon crumble, served with vanilla ice cream.